

CASES IN CYBERSECURITY 2

Advanced Cases in Assurance Services (ACCTG 521)
Class 17 | MPAcc class of 2026

Agenda

Review and look ahead (workshop classes)

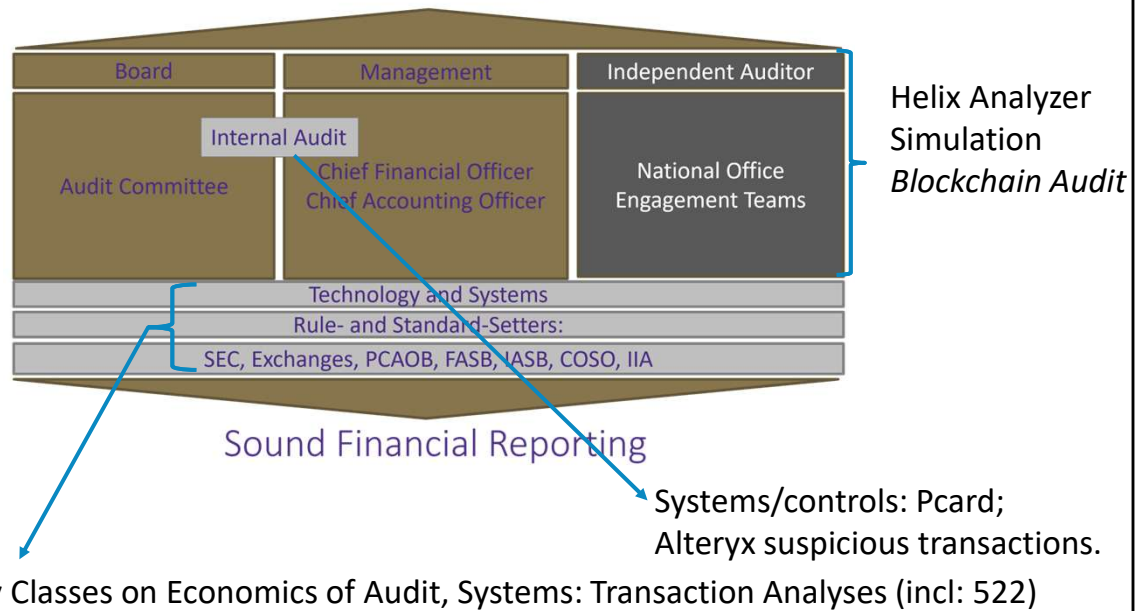
Deliverable

Lab:

- Escape the Digital Dungeon Case

Review

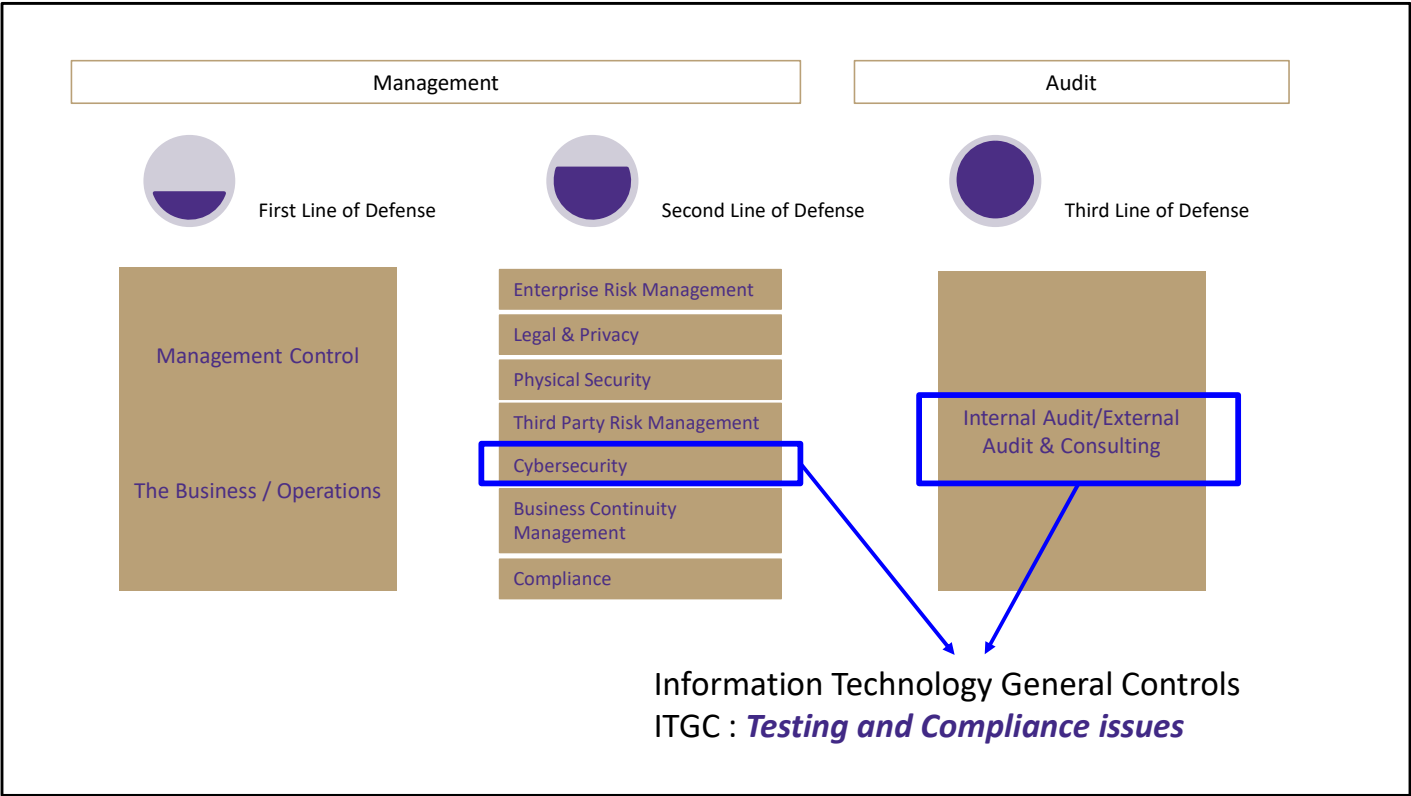
Effective Financial Reporting Governance

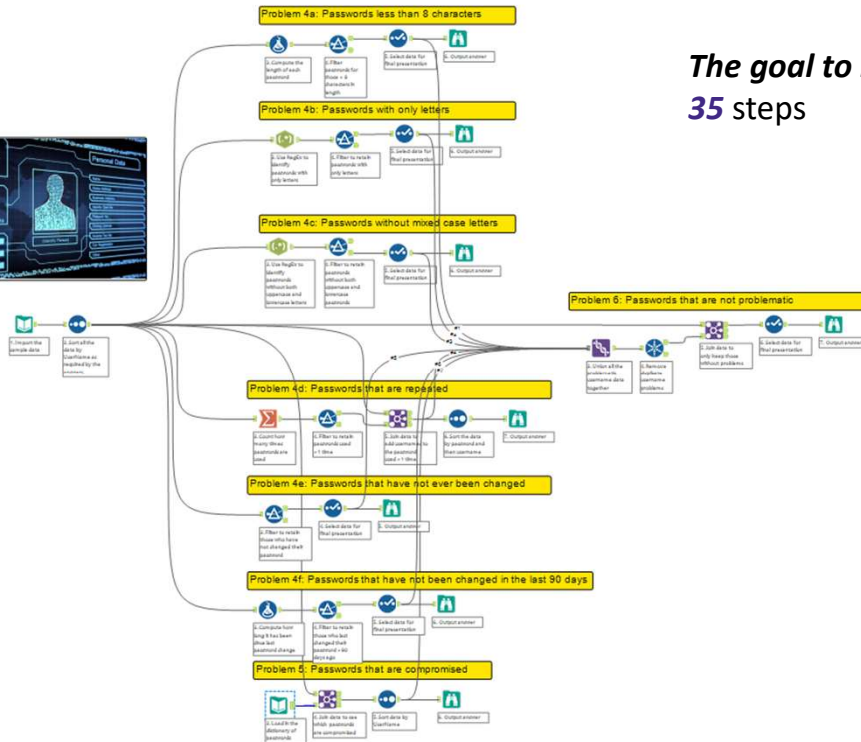


Bob – Course Overview (3 legged stool)

Christina – Refer to Syllabus and go into depth on professionalism piece

See if class has any questions on syllabus





The goal to beat:
35 steps

Interface Designer - Find and Replace

Browse (8) - Configuration

6 records displayed, 14 fields, 1,775 bytes

Profile

- Billings 1
- Ellis 1
- Lugo 1
- Redden 1
- 1 more >

Department

- Accounting and Finance 6

Password

- &K[EN^M5nVde7 1
- 8\$!y-KDnKxj3 1
- >/CR6Uxm%EA 1
- yC4Fvj=qb>NS 1
- (&TgVj3*dbZUWP? 1
- 1 more >

PwdChangeDate

- 2022-03-06 3
- 2022-03-08 1
- 2022-03-15 1
- 2022-03-29 1

PwdInitial

Results - Browse (8) - Input

14 of 14 Fields | Cell Viewer | 6 records displayed, 1,775 bytes

Record	UserName	LastName	FirstName	Department	Password	PwdChangeDate	PwdInitial	duplicatePwd	bool_pwdLength	pwdLetterOnly	pwdUpperandLower	pwdNoPunctuation	diff_days	over90days
1	BarclayH562	Hannah	Barclay	Accounting and Finance	&K[EN^M5nVde7	2022-03-29	0	False	False	False	True	False	36	False
2	BillingsN726	Naida	Billings	Accounting and Finance	>/CR6Uxm%EA	2022-03-06	0	False	False	False	True	False	59	False
3	ShafferN129	Nicky	Shaffer	Accounting and Finance	(&TgVj3*dbZUWP?	2022-03-15	0	False	False	False	True	False	50	False
4	ReddenZ703	Zoe	Redden	Accounting and Finance	yC4Fvj=qb>NS	2022-03-08	0	False	False	False	True	False	57	False
5	LugoK615	Kendal	Lugo	Accounting and Finance	8\$!y-KDnKxj3	2022-03-06	0	False	False	False	True	False	59	False
6	EllisM154	Moesha	Ellis	Accounting and Finance	J/*IGAT	2022-03-06	0	False	False	False	True	False	59	False

Workflow Diagram:

```

graph LR
    Input[Input] --> Password[Password - Ascending]
    Password --> Filter1[Filter: (Row-1:Password) OR (Row+1:Password) OR (Row-1:Password) OR (Row+1:Password)]
    Filter1 --> Filter2[Filter: bool_pwdLength (duplicatePwd) = IF Length OR over90days]
    Filter2 --> Filter3[Filter: (Password) <= 0]
    Filter3 --> Filter4[Filter: THEN 1 ELSE 0]
    Filter4 --> Filter5[Filter: ENDIF]
    Filter5 --> Filter6[Filter: (pwdLetterOnly) or (pwdLetterOnly) or (pwdLetterOnly)]
    Filter6 --> Output[Output]
  
```

Workflow Diagram Details:

- Input
- Filter: Password - Ascending
- Filter: (Row-1:Password) OR (Row+1:Password) OR (Row-1:Password) OR (Row+1:Password)
- Filter: bool_pwdLength (duplicatePwd) = IF Length OR over90days
- Filter: (Password) <= 0
- Filter: THEN 1 ELSE 0
- Filter: ENDIF
- Filter: (pwdLetterOnly) or (pwdLetterOnly) or (pwdLetterOnly)
- Output

Key Cybersecurity Take-Aways

- Many passwords are not secure, IT Controls need to work on password policy.
- Today: Software patching can prevent malware injections.

Look ahead (workshop classes)

Workshop classes

Goal facilitate completion of the final project.

Class #	ACCTG 522	ACCTG521
18	11/24 Workshop	11/25 Blockchain (individual)
19	11/26 Thanksgiving break	11/27 Thanksgiving break
20	12/2 Workshop	12/2 Workshop *Audit Plan Deadline
21	12/3 Workshop *Final Project Due by 11:59PM	12/4 Final Presentations

Workshops do not require classroom presence, if teams want to meet remotely or in breakout rooms, let me know.

Workshop focus

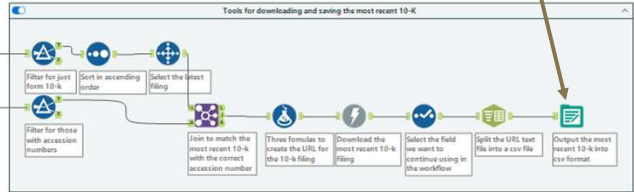
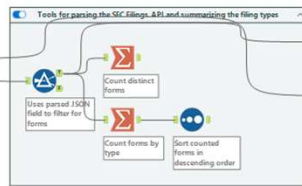
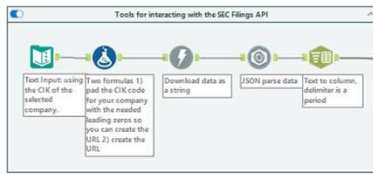
<https://forms.office.com/r/mTpGsZX5hu>

Full text filing (text, tables, XBRL schema and more)

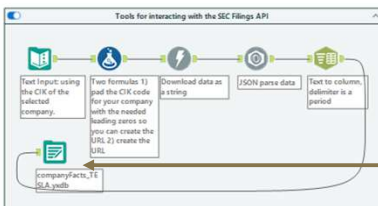
The steps here include using the input text tool to provide the URL for the download tool to connect to the API, the JSON data returned by the download tool is the parsed using the JSON tool which can then be deserialized further using the text to columns tool.

The steps here are to identify all of the components needed to answer the questions about the filings (the filter and summarize tools).

The tools in this step are used to identify the most recent 10-K filing components. First the components to construct the URL which can be used to download the most recent 10-K filing uses the filter tool, the sort and select records tools. Then using a join tool, and a formula tool, to construct the URL, the download tool is used to interact with the SEC's website to download the most recent 10-K. A select tool and text to columns tool are then used to create a csv formatted 10-K which is saved using the output data tool.



Use this step first to download the SEC API data from company facts. It will provide output in the form of a json file with all of the XBRL data stored and indexed with some summary filings information (e.g., form filed 10-K/Q and dates).



JSON file of XBRL (largely numeric)

Final Project Templates

Data Source Appendix

Goal: Have a summary Table documenting the sources of data used and where the data was used.

Appendix 1
Data Sources

Data Source	Used for
SEC-API-Company-Facts	XBRL-extracted-data-used-as-input-to-eVal-Historical-Financial-Statement-data
10-K:Item-7-MD&A	Additional-risk-analysis-around-changes-in-sentiment
10-K:Non-GAAP-reconciliation	Assessing-ratio-sensitivity-to-non-GAAP-ratios-and-alternative-metrics

¶

..... Page Break ¶

Final Project Templates

Analytics Enhancement Appendix

Goal: Have a summary Table documenting the sources analytical methods used and where in the Financial Statement Analysis they were incorporated.

¶

Appendix 2¶
Summary of Analytical Enhancements to FSA¶

¶

Enhancement¶	Used to¶	¶
Disaggregate forecasting of revenue¶	Separate individual revenue streams to better predict future Revenue¶	¶
¶	¶	¶
¶	¶	¶
¶	¶	¶

¶

.....Page Break.....¶

Final Project Templates

XBRL Mapping to eVal

Goal: Have a summary Table documenting the mapping of disaggregated tags to eVal line items.

¶

Appendix 3 (optional)¶
XBRL Mapping to eVal¶

eVal Line Item¶	XBRL Inputs¶
<i>Income Statement¶</i>	
Sales (Net)¶	¶
Cost of Goods Sold¶	¶
Gross Profit¶	(Calculated)¶
R&D Expense¶	¶
SG&A Expense¶	¶
EBITDA¶	(Calculated)¶
Depreciation & Amortization¶	¶
EBIT¶	(Calculated)¶
Net Interest Expense¶	¶
Non-Operating Income (Loss)¶	¶
EBT¶	(Calculated)¶
Income Taxes¶	¶
Other Income (Loss)¶	¶
Net Income Before Ext. Items¶	¶
Ext. Items & Disc. Ops.¶	¶
Noncontrolling Interest in Earnings¶	¶
Preferred Dividends¶	¶
Net Income (available to common)¶	(Calculated)¶

¶

¶

Alteryx Challenge 11/24

Two required submission portals with multiple files:

- Individual Case Submissions: **Alteryx Workflows** -> all *completed* SparkEd (up to 6 files) and *complete/incomplete* Challenge (3 files)
- Individual Case Submissions: **Alteryx Challenge Memo** -> memo about the three Alteryx challenges in 11/16 class (with Nicole as guest).

Agenda

Review

Deliverable

Lab:

- Escape the Digital Dungeon Case

Escape from the Digital Dungeon

Ernst & Young
FOUNDATION™
Academic Resource Center

Provided by the EYARC



Enter your Alteryx generated code

A response will appear,
and if you are correct
the “unlock code” copy
it and paste it into a
comment on your
workflow along with
your name.

<https://eyarc.site/cyber/index.html>

Deliverable

Name: Asher Curtis
Unlock Code: _____

Comment tool with name and code

Rest of your workflow



**Save workflow as a packaged workflow

FOSTER
Master of
Professional Accounting

Deliverable

Your final Alteryx workflow, as a packaged workbook.

The screenshot illustrates the process of exporting an Alteryx workflow as a packaged workbook. It shows the Alteryx interface with the 'Options' menu open, highlighting the 'Export Workflow' option (labeled 1). The 'Export Workflow' dialog box is also shown, displaying the 'Export Settings' section where the workflow name is 'pure_oils_submission_combiner.yxmd' and the package location is 'U:\my_preclass_work\ACCTG_521\case05_PureOils\pure_oils_submission_combiner.yxmd' (labeled 2). The 'Group Assets by Tool' checkbox is checked, and the output path is 'C:\GitHubConnections\mylocalhelpers\pureOils\pure_oils_submission_combiner.yxmd' (labeled 3, noting the file extension is '.yxzp'). A green status bar at the bottom confirms the successful export: 'Workflow exported successfully. Package written to U:\my_preclass_work\ACCTG_521\case05_PureOils\pure_oils_submission_combiner.yxzp'.

1

2

3 (note file ext is ".yxzp")

✓ Workflow exported successfully. Package written to U:\my_preclass_work\ACCTG_521\case05_PureOils\pure_oils_submission_combiner.yxzp

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Today's Challenge

Add a **macro component** as part of the workflow solution.

- Must be a part of the workflow done more than once.

Case Cryptic Clue:

Search every person's files, Search high and low!
I used just ~~one person's~~ two peoples' files For
my password bold. Password? You say. What is
it? You ask. Find all the digits Sum them up! It's
the only way To avoid bad luck. Time is ticking ...
Rumpelstiltskin

Remainder of case gives hints about the workflow

Read through including data description

Think about starting point of analysis, use **Required** section to help set-up workflow order.

Hints includes reminders of prior cases.

Hints

Hints

This case does not require using macros, but it can be solved by using macros.

The Input Data tool in Alteryx can be used to read multiple files at one time. You can learn more at <https://help.alteryx.com/20223/designer/input-data-tool>.

When importing data into Alteryx, you do not always have to have a “neat and tidy” import if you don’t need all the data. You can import, clean it and then structure your data as appropriate.

The RegEx tool is a powerful way to parse data. If you use the Tokenize output method, you can split data into rows.

Thank you!

